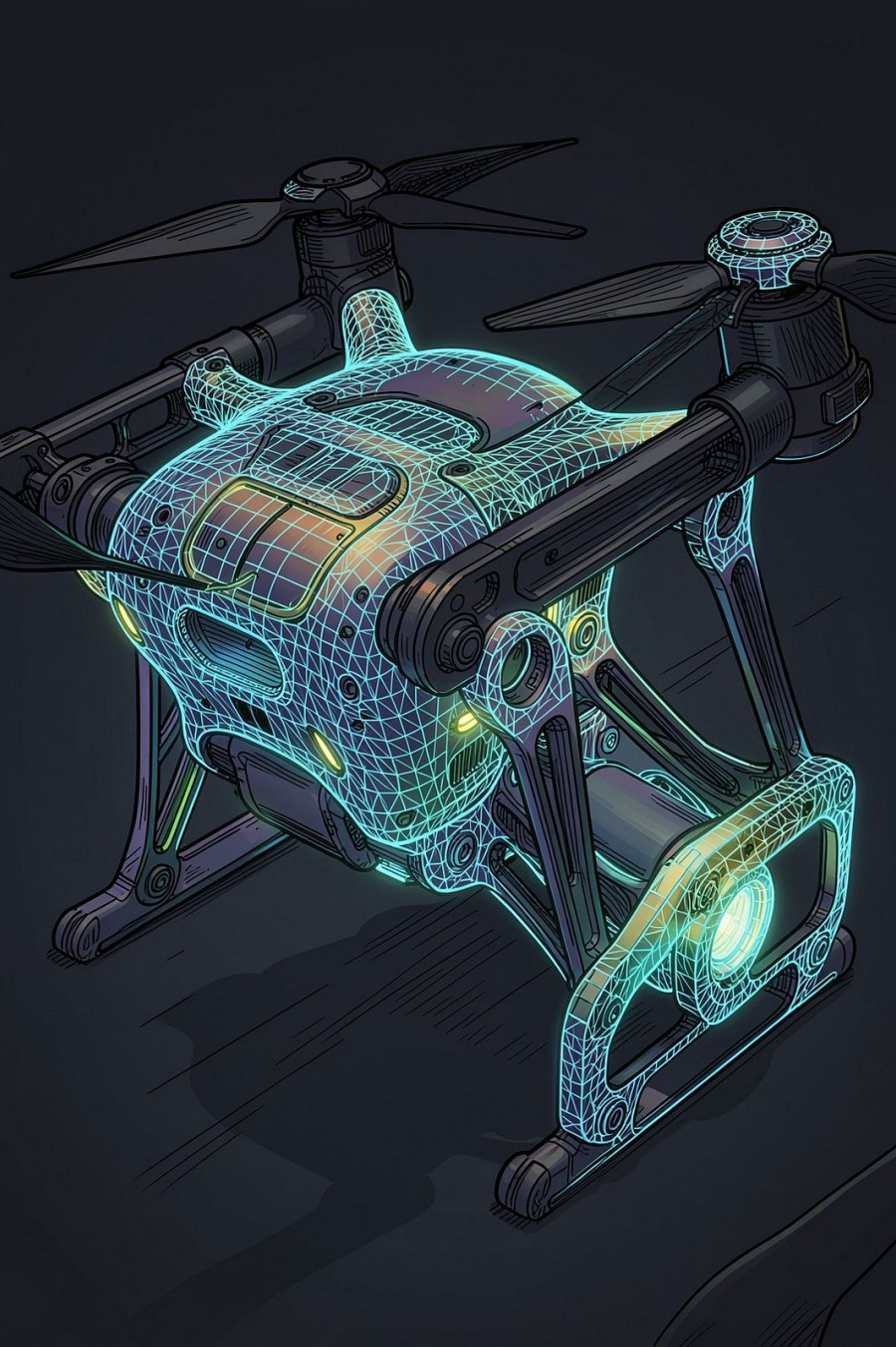




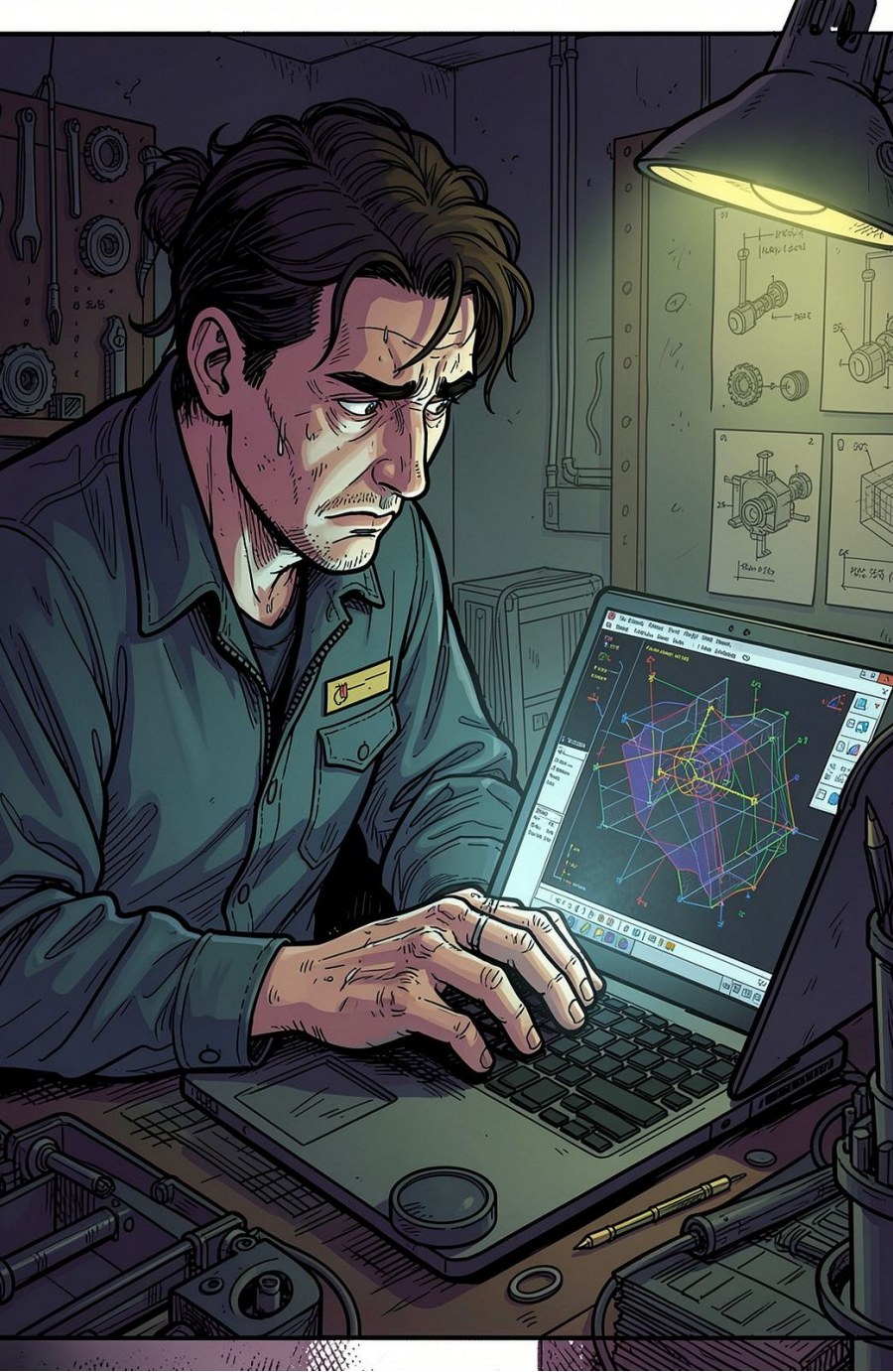
Generative CAD Assistant

On-Device Topological Optimization for Snapdragon-Powered HP PCs

AI-POWERED DESIGN

SNAPDRAGON OPTIMIZED

Made with **GAMMA**



The Hardware Bottleneck Is Real

Advanced 3D modeling and structural simulation in tools like **SolidWorks** and **Ansys** demand expensive discrete GPUs or cloud processing. Engineers and students are routinely delayed — not by design complexity, but by hardware limitations.

Cost Barrier

Workstation-grade GPUs are prohibitively expensive for students and small teams.

Cloud Dependency

Offloading to the cloud introduces latency and exposes sensitive IP.

Iteration Delay

Complex topological optimizations stall workflows for hours on standard laptops.

Where the Problem Hits Hardest



Lessons from the Field

As a founding member of the **Association of Mechanical Engineers**, organizing technical competitions like the **IGNITION CAD Challenge** revealed a consistent pattern: participants with strong design ideas were bottlenecked by hardware, not skill.

- ❑ Complex topological optimization runs were abandoned or simplified — not due to lack of talent, but lack of compute.



Introducing the Generative CAD Assistant

An on-device AI tool that integrates into existing mechanical engineering workflows — analyzing exported 3D meshes and geometries to **instantly suggest weight-saving topological optimizations and structural reinforcements**.

→ Analyze

Ingests exported 3D meshes and CAD geometries directly.

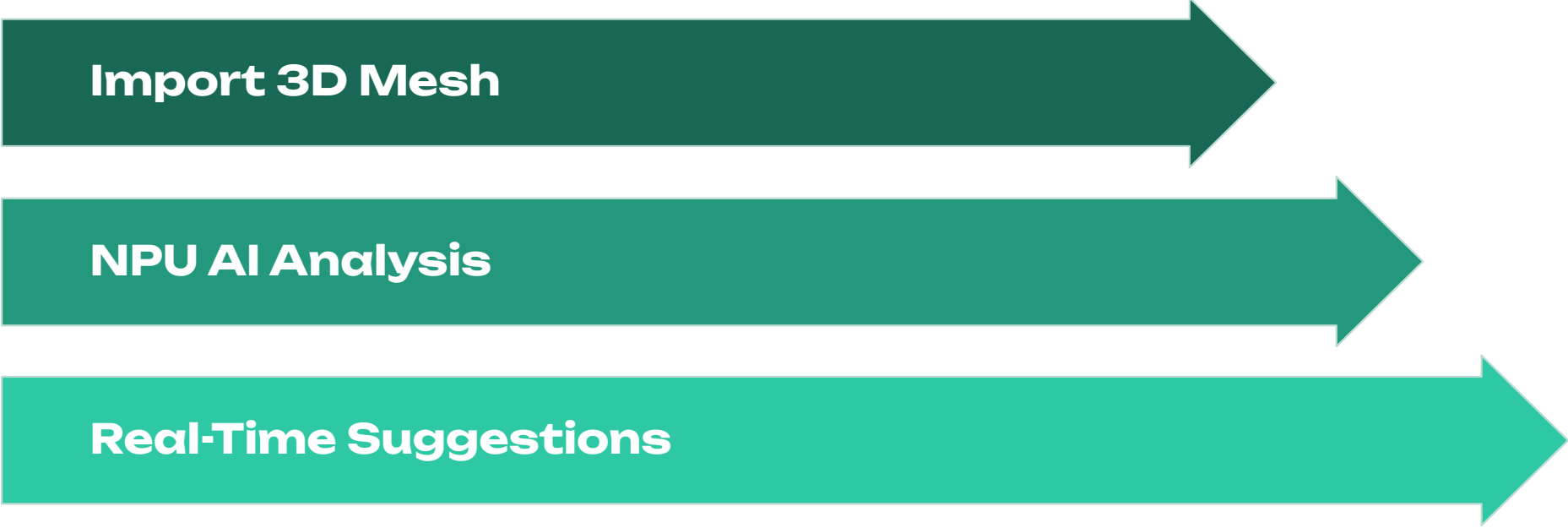
→ Reason

AI evaluates structural integrity and identifies optimization opportunities.

→ Suggest

Delivers real-time, actionable design feedback during the design phase.

Architecture: Shifting Intelligence to the Edge



Import 3D Mesh

NPU AI Analysis

Real-Time Suggestions

By moving the reasoning workload from the cloud to the local machine, the Generative CAD Assistant eliminates iteration lag and keeps proprietary designs entirely on-device.

Built for Snapdragon: Heterogeneous Compute

The assistant is explicitly optimized for **Snapdragon-powered HP PCs**, leveraging Qualcomm's advanced heterogeneous compute architecture to distribute workloads intelligently.

Qualcomm AI Hub Integration

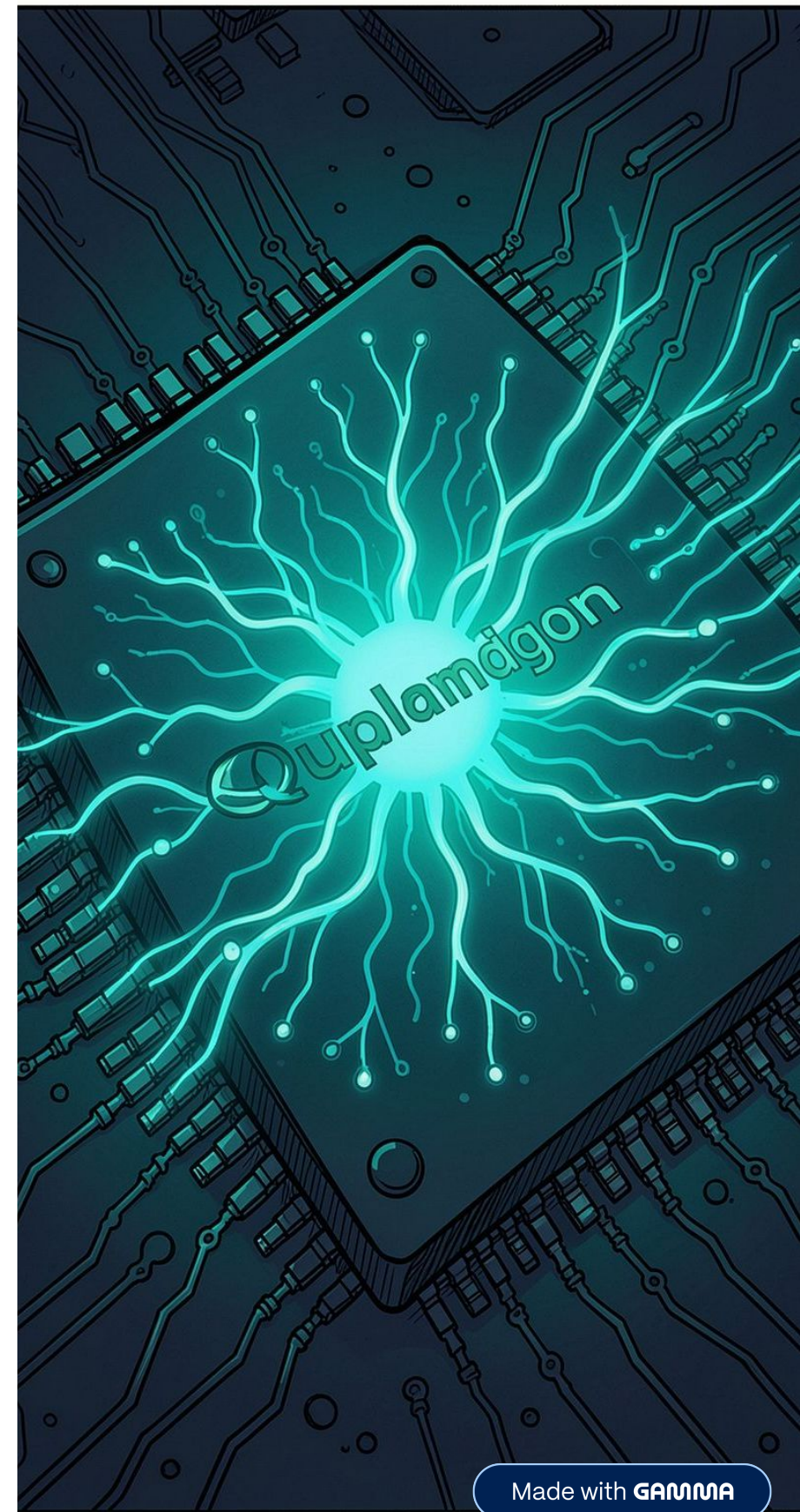
Pre-quantized geometric deep learning and computer vision models from the Qualcomm AI Hub ensure compatibility and performance out of the box.

NPU Offloading

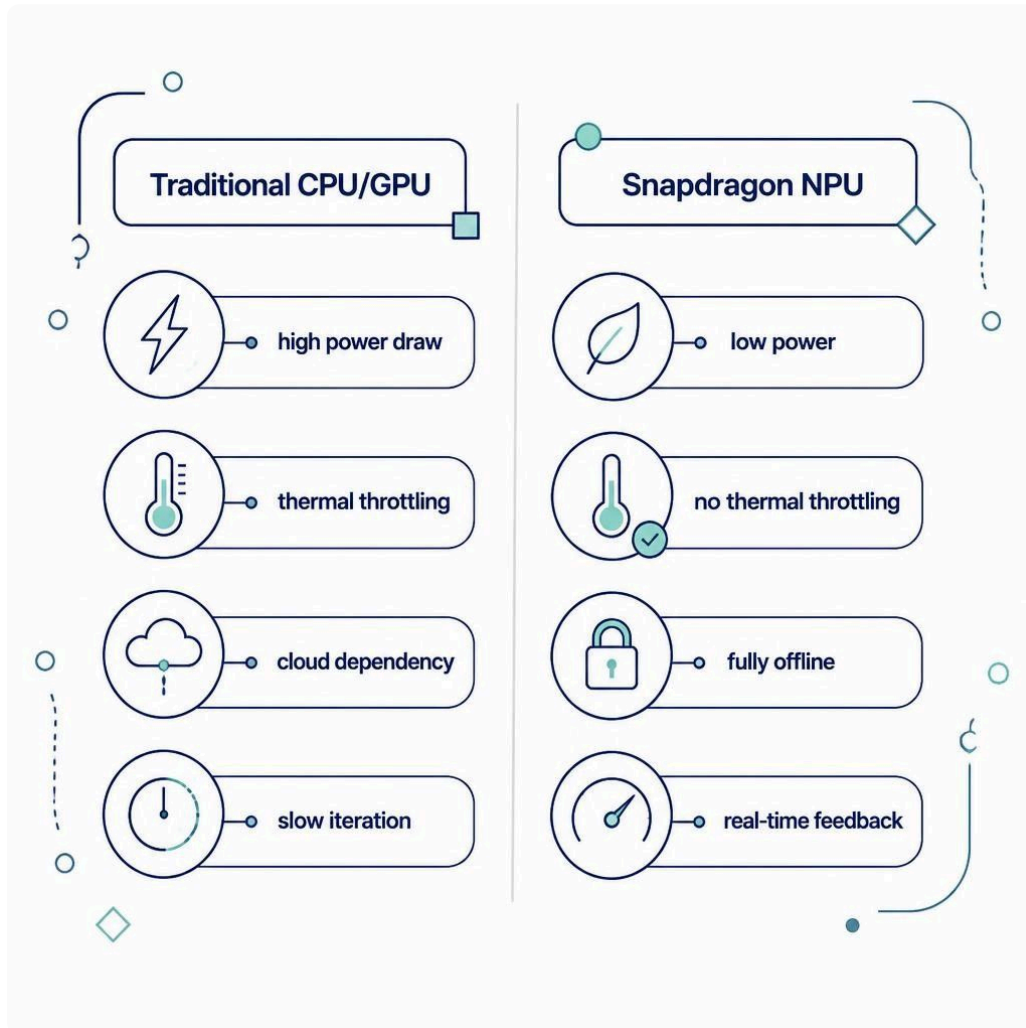
Heavy inferencing tasks — spatial awareness, structural integrity analysis — are offloaded to the Snapdragon NPU, freeing CPU and GPU for CAD rendering.

Thermal & Power Efficiency

NPU-driven AI workloads operate within the power envelope of thin-and-light Snapdragon PCs, avoiding thermal throttling entirely.



NPU vs. Traditional Compute: The Efficiency Gap



Why NPU Changes Everything

Running AI inferencing on the Snapdragon NPU instead of the CPU or GPU means:

- CAD rendering performance stays uncompromised
- No thermal throttling on thin-and-light hardware
- Power consumption stays within laptop envelope
- Zero dependency on internet connectivity

Data Privacy by Architecture

Running the Generative CAD Assistant **completely offline** ensures that sensitive intellectual property — proprietary drone designs, robotic mechanisms, and unreleased product geometries — **never leaves the device**.

No Cloud Upload

Geometry and design data never traverses a network.

No Third-Party Exposure

Proprietary IP stays under the engineer's full control.

Compliance-Ready

Meets enterprise and defense-grade data governance requirements.



Democratizing Generative Design

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Cloud Dependency

Fully offline operation

100%

On-Device AI

All inferencing local

~OW

GPU Overhead

NPU handles AI load

Advanced Tools, Accessible Hardware

Generative design has historically been reserved for well-funded teams with workstation budgets. The Generative CAD Assistant changes that equation — enabling **students, startups, and field engineers** to run sophisticated topological optimization on lightweight Snapdragon HP PCs.

✓ Same capability. Fraction of the hardware cost. Zero compromise on privacy.

The Path Forward

The Generative CAD Assistant represents a paradigm shift — from hardware-constrained, cloud-dependent workflows to **intelligent, private, on-device design acceleration**.

O1

Integrate Qualcomm AI Hub Models

Embed pre-quantized geometric deep learning models for Snapdragon NPU.

O2

Build CAD Plugin Interface

Develop seamless integration with SolidWorks, Ansys, and open CAD platforms.

O3

Validate on Snapdragon HP PCs

Benchmark thermal performance, inference latency, and optimization accuracy.

O4

Deploy to Engineers & Students

Release to the mechanical engineering community — starting with IGNITION participants.

